

PROJECT 3

SQL ANALYSIS REPORT

This project focused on analyzing the Online Retail Dataset using Microsoft SQL Server and SQL Server Management Studio. The objective of the project was to apply SQL querying, data cleaning, filtering, aggregation, and reporting techniques to extract meaningful business insights from retail transaction data.

SQL TECHNIQUES APPLIED

The project demonstrated practical use of core SQL concepts and analytical functions, including:

SELECT statements for data extraction

WHERE clause for filtering records

ORDER BY for sorting results

GROUP BY for aggregation

HAVING clause for filtering grouped data

Aggregate Functions: SUM ()

AVG ()

COUNT ()

Additional SQL operations included data updates, date extraction functions, and conditional filtering for business analysis.

KEY FINDINGS

OVERALL PERFORMANCE

1. Total Orders: 1,200
2. Total Revenue: 1,264,761.96
3. Average Order Value: 1,053.97
4. Average Quantity Ordered: 2 items
5. Cancelled Orders: 251

CUSTOMER AND PAYMENT INSIGHTS

1. Most Used Payment Method: Online Payments (259 transactions)
2. Top Payment Method: Credit Card (\$263,847.63)
3. Top Referral Source: Instagram (260 referrals)
4. Top Referral Source: Instagram (\$275,285.45)

PRODUCT PERFORMANCE

The analysis identified strong-performing products across the dataset:

Top 3 Products by Revenue:

1. Chair — \$195,620.11
2. Printer — \$195,612.61
3. Laptop — \$192,126.56

Products with sales greater than 2000 included Chair, Desk, Laptop, Monitor, Phone, Printer, and Tablet.

Product with the highest individual total price:

1. Tablet
2. Coupon Code: SAVE10
3. Total Price: 3,456.40

Product with the lowest recorded price was:

1. Phone
2. Referral Source: Email
3. Total Price: 11.39

REVENUE ANALYSIS

REVENUE BY PAYMENT METHOD

S/N	Payment Method	Total Revenue
1	Credit Card	\$263,847.63
2	Online	\$262,442.94
3	Cash	\$259,786.29
4	Gift Card	\$246,323.92
5	Debit Crad	\$232,361.18

REVENUE BY REFERRAL SOURCE

S/N	Referral Source	Total Revenue
1	Instagram	\$275,285.45
2	Email	\$261,808.55
3	Google	\$250,441.48
4	Facebook	\$250,410.90
5	Referral	\$226,815.58

CONCLUSION

This project demonstrated practical SQL and data analysis skills using a real-world retail dataset. SQL queries were effectively used to clean, organize, analyze, and summarize transactional data into actionable business insights. The analysis provided visibility into customer behavior, product performance, payment trends, and revenue distribution. Overall, the project strengthened foundational SQL skills and showcased the ability to transform raw data into meaningful business intelligence suitable for decision-making and reporting purposes.